mechanics, it was “the greatest intellectual thrill” of his life.8 Cumrun Vafa, a leading string theorist from Harvard University, has said that “string theory is definitely revealing the deepest understanding of the universe which we have ever had.”9 And Nobel Prize-winner Murray Gell-Mann has said that string theory is “a fantastic thing” and that he expects that some version of string theory will someday be the theory of the whole world.10

As you can see, the debate is fueled in part by physics and in part by distinct philosophies about how physics should be done. The “traditionalists” want theoretical work to be closely tied to experimental observation, largely in the successful research mold of the last few centuries. But others think that we are ready to tackle questions that are beyond our present technological ability to test directly.

Different philosophies notwithstanding, during the past decade much of the criticism of string theory has subsided. Glashow attributes this to two things. First, he notes that in the mid-1980s,

String theorists were enthusiastically and exuberantly proclaiming that they would shortly answer all questions in physics. As they are now more prudent with their enthusiasm, much of my criticism in the 1980s is no longer that relevant.11

Second, he also points out,

We non-string theorists have not made any progress whatsoever in the last decade. So the argument that string theory is the only game in town is a very strong and powerful one. There are questions that will not be answered in the framework of conventional quantum field theory. That much is clear. They may be answered by something else, and the only something else I know of is string theory.12

Georgi reflects back on the 1980s in much the same way:

At various times in its early history, string theory has gotten oversold. In the intervening years I have found that some of the ideas of string theory have led to interesting ways of thinking about physics which have been useful to me in my own work. I am much happier now to see people spending their time on string theory since I can now see how something useful will come out of it.13

Theorist David Gross, a leader in both conventional and string physics, has eloquently summed up the situation in the following way: